

MARIAZELL, Austria

WMO: 111720

Lat: 47.767N

Lon: 15.317E

Elev: 2871

StdP: 13.23

Time Zone: 1.00 (EUC)

Period: 83-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.6	(c) 8.6	(d) -5.2	(e) 4.6	(f) 10.5	(g) 0.6	(h) 6.3	(i) 11.3	(j) 28.7	(k) 35.0	(l) 24.8	(m) 34.2	(n) 0.7	(o) 120	(p) 0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.3	(c) 80.3	(d) 60.6	(e) 76.8	(f) 59.6	(g) 73.9	(h) 58.4	(i) 62.9	(j) 74.6	(k) 61.3	(l) 72.8	(m) 60.0	(n) 70.7	(o) 11.4	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.0	82.9	66.8	57.2	77.8	64.8	55.7	73.3	63.5	29.9	74.2	28.7	73.0	27.7	71.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.9	(b) 22.6	(c) 20.8	(e) -4.3	(f) 85.8	(g) 8.2	(h) 2.8	(i) -10.2	(j) 87.9	(k) -15.0	(l) 89.6	(m) -19.6	(n) 91.2	(o) -25.5	(p) 93.2		
(4) 24.9	(5) 22.6	(6) 20.8	(7) -4.9	(8) 66.7	(9) 8.1	(10) 2.7	(11) -10.7	(12) 68.6	(13) -15.4	(14) 70.1	(15) -20.0	(16) 71.6	(17) -25.8	(18) 73.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.0	27.6	30.1	34.7	42.1	52.1	56.0	60.2	60.4	52.8	46.4	35.6	28.8
	DBStd	13.90	8.58	8.51	7.86	7.13	6.73	6.37	5.64	5.94	6.31	7.31	8.00	8.18
	HDD50	3350	696	557	477	255	55	14	2	4	42	157	435	656
	HDD65	7725	1161	977	941	688	401	278	168	163	367	578	882	1121
	CDD50	1152	0	0	1	17	120	195	317	328	126	45	3	0
	CDD65	50	0	0	0	0	1	9	18	21	1	0	0	0
	CDH74	615	0	0	0	0	13	106	227	256	13	0	0	0
	CDH80	93	0	0	0	0	0	13	35	45	0	0	0	0
(14) Wind	WSAvg	5.2	4.4	5.3	5.5	6.1	5.6	5.1	5.1	4.7	4.7	6.0	5.2	4.6
(15) Precipitation	PrecAvg	43.5	2.6	2.5	3.4	2.8	4.3	5.1	5.6	4.6	4.1	2.9	3.0	2.8
	PrecMax	55.8	6.9	5.4	7.4	4.8	11.5	13.3	14.0	10.7	11.6	7.3	6.0	5.4
	PrecMin	30.0	0.2	0.6	1.4	0.6	1.5	1.8	1.6	1.4	1.4	0.2	0.0	0.8
	PrecStd	6.2	1.6	1.4	1.7	1.2	2.2	2.3	2.9	2.1	2.2	1.6	1.4	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.3	55.2	61.5	68.2	75.7	82.3	84.5	84.7	76.2	70.2	59.9	50.4
		MCWB	38.8	41.7	46.0	51.3	56.8	60.8	63.0	61.5	58.6	53.9	48.3	41.1
	2%	DB	45.1	49.0	55.8	63.4	72.4	77.3	80.1	80.3	72.1	65.5	54.9	46.2
		MCWB	37.4	39.8	42.9	48.8	55.1	59.8	61.3	60.3	57.1	52.6	45.3	39.4
	5%	DB	42.0	45.4	51.1	59.2	68.6	73.3	76.5	76.4	68.4	61.8	51.3	43.2
		MCWB	35.8	37.6	40.7	46.6	53.6	58.2	59.9	59.7	55.9	51.0	43.0	37.9
	10%	DB	38.9	41.7	46.5	55.1	65.0	69.3	72.9	73.0	65.0	58.3	47.9	39.7
		MCWB	34.3	35.7	38.9	44.7	52.1	56.1	58.7	58.6	54.5	49.5	41.6	35.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.4	43.2	47.1	53.5	59.7	64.3	66.6	64.6	61.1	56.4	49.6	43.1
		MCDB	45.7	51.2	58.9	65.3	70.4	76.2	78.6	76.2	69.8	65.8	57.5	48.7
	2%	WB	38.4	40.7	44.0	50.3	56.9	61.1	63.2	62.6	59.1	54.1	46.5	40.5
		MCDB	43.8	47.3	53.4	60.5	68.9	73.5	75.0	74.4	69.2	62.5	53.6	44.5
	5%	WB	36.6	38.4	41.8	47.6	54.9	59.3	61.6	61.0	57.4	52.3	44.2	38.4
		MCDB	41.1	44.2	49.2	56.8	65.6	70.1	72.8	72.8	65.5	59.6	50.3	42.7
	10%	WB	34.5	36.2	39.6	45.3	53.0	57.3	60.0	59.6	55.5	50.4	41.9	36.2
		MCDB	38.4	41.0	45.5	53.4	63.0	67.4	70.2	70.5	63.2	57.1	47.0	39.5
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	14.4	14.3	15.3	17.3	16.6	17.3	18.0	16.0	15.5	12.3	11.0
		MCDBR	16.2	20.5	24.3	24.0	24.2	25.0	26.0	26.2	24.0	23.0	17.9	14.8
		MCWBR	10.4	12.3	13.3	12.1	11.4	10.4	11.0	10.1	11.3	11.9	10.7	10.1
	5% WB	MCDBR	14.3	17.7	21.4	20.3	21.7	22.1	21.6	22.1	20.5	18.6	15.4	12.6
		MCWBR	10.0	11.5	12.7	11.1	11.0	10.2	10.2	9.7	10.3	10.8	10.1	9.5
(40) Clear-Sky Solar Irradiance	taub		0.283	0.311	0.351	0.391	0.395	0.400	0.405	0.401	0.373	0.342	0.313	0.283
	taud		2.405	2.350	2.301	2.250	2.280	2.299	2.286	2.313	2.366	2.435	2.466	2.450
	Ebn at Noon		254	269	274	272	275	274	271	266	264	255	240	241
	Edh at Noon		21	28	35	40	41	40	40	37	32	26	20	18
(44) All-Sky Solar Radiation	RadAvg		363	639	968	1322	1530	1671	1653	1478	1066	716	392	298
	RadStd		43	80	111	145	175	135	154	165	136	71	52	40

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only		(d) +1.44	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47) Regional (105 neighbors)		+0.94	N/A	N/A	+1.49	+0.81	+0.72	-187	-250	+151	+79

Nomenclature: See separate page